Ministry of Education

Umm Al-Qura University

Collage of computer and information system

وزارة التعليم

جامعة ام القرى

كلية الحاسب الالي ونظم المعلومات

SUMMER TRAINING REPORT

AT:

Medical Fakeeh

DURATION:

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SUBMITTED BY:

Hala Mohammed Jalal

[redacted]

SUPERVISED BY:

Dr. Hana Abdullah Alghamdi

Department of Computer Science and Artificial Intelligence

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Introduction

Summer Training Requirements and Plan Selection

The summer training program is an essential requirement for completing my degree in computer science. It serves as an opportunity for students to apply theoretical knowledge in a real-world setting, gain hands-on experience, and develop professional skills that are crucial for future career success. The training is designed to immerse students in a professional environment where they can contribute to projects, collaborate with industry professionals, and tackle real-life challenges that arise in their field of study.

When selecting my summer training plan, I aimed to align it with my career aspirations and academic interests. The choice was influenced by my desire to deepen my understanding of how technology can be applied in healthcare, a sector that increasingly relies on innovative IT solutions. I opted for training at Medical Fakeeh because of the institution's reputation for integrating advanced technology with healthcare services, which provided a unique opportunity to explore the intersection of these two fields.

About Medical Fakeeh

History

Medical Fakeeh has a storied history that began in the 1970s as Saudi Airlines Medical Services, initially established to serve the healthcare needs of Saudi Arabian Airlines employees and their families. Over the decades, the organization evolved, expanding its services and infrastructure to meet the growing demands of its clientele. This evolution culminated in a significant transformation when the organization rebranded as Medical Fakeeh, marking a new chapter in its journey as a leading healthcare provider in Saudi Arabia.

Achievements

Throughout its history, Medical Fakeeh has achieved numerous milestones that underscore its commitment to healthcare excellence. Some of the key achievements include:

- ***Expansion of Services:*** From a small medical unit, Medical Fakeeh has grown into a comprehensive healthcare provider with state-of-the-art clinics and medical centers equipped with the latest medical technology.
- ***Partnership with Fakeeh Care***: The association with the esteemed Fakeeh Care group has further enhanced the organization’s capabilities, bringing in advanced healthcare solutions and a patient-centered approach.
- ***Community Impact:*** Medical Fakeeh has consistently contributed to the well-being of the community, extending its services beyond the initial scope of Saudi Arabian Airlines to serve a broader population.

Services

Medical Fakeeh offers a wide range of services designed to meet the diverse healthcare needs of its patients. These services include:

- ***Comprehensive Healthcare***: From primary care to specialized medical treatments, Medical Fakeeh provides a full spectrum of healthcare services, ensuring that patients receive the best possible care.
- ***Advanced Medical Facilities:*** The clinics and medical centers are equipped with cutting-edge technology, enabling accurate diagnosis and effective treatment for various medical conditions.
- ***Qualified and Compassionate Staff:*** The healthcare professionals at Medical Fakeeh are not only highly qualified but also deeply committed to providing compassionate and personalized care to every patient.
- ***Innovative Health Solutions:*** Embracing the latest in digital health services, telemedicine, and advanced patient care technologies, Medical Fakeeh continually strives to enhance the patient experience and improve healthcare outcomes.



Training Plan

الأسبوع Week	المشاريع / المهام Tasks / Projects
1	The first week will be an orientation about the organization.
2	how to Troubleshoot the issues that related to IT.
3	introduction to Active directory (AD) how to create\modify\delete users and how to join\unjoin computer in the domain.
4	will be for HIS (hospital information system) how to fix all the issues in HIS system.
5	How to troubleshoot network issues.
6	Understanding Development Processes and Tools
7	Technical Skills Development
8	Project Management and Leadership

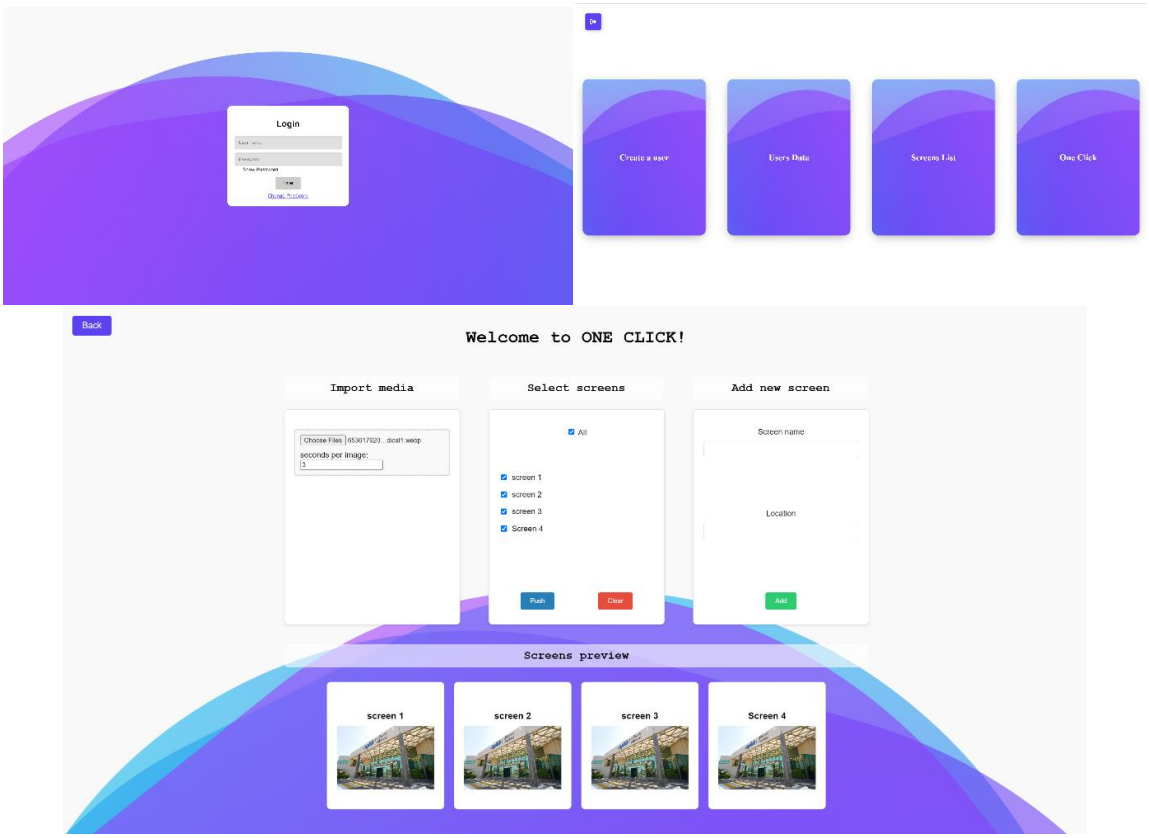
Training Overview

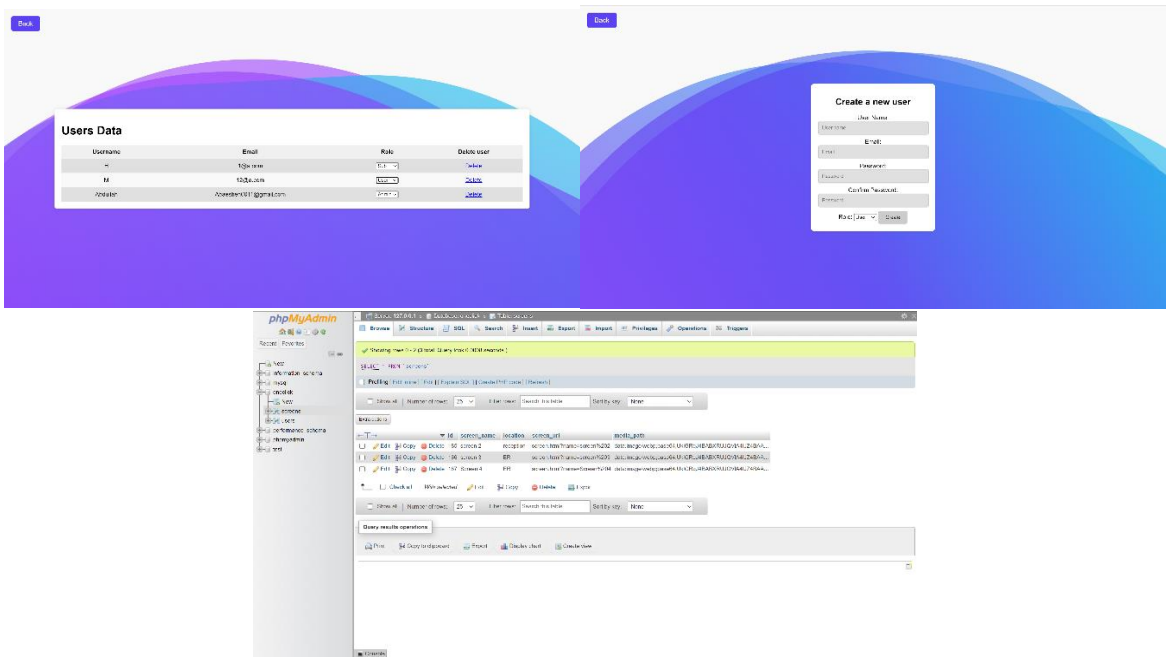
During my summer training at Medical Fakeeh, I had the opportunity to work on several intriguing projects, encounter interesting events, and develop valuable skills that significantly contributed to my professional growth.

Interesting Projects

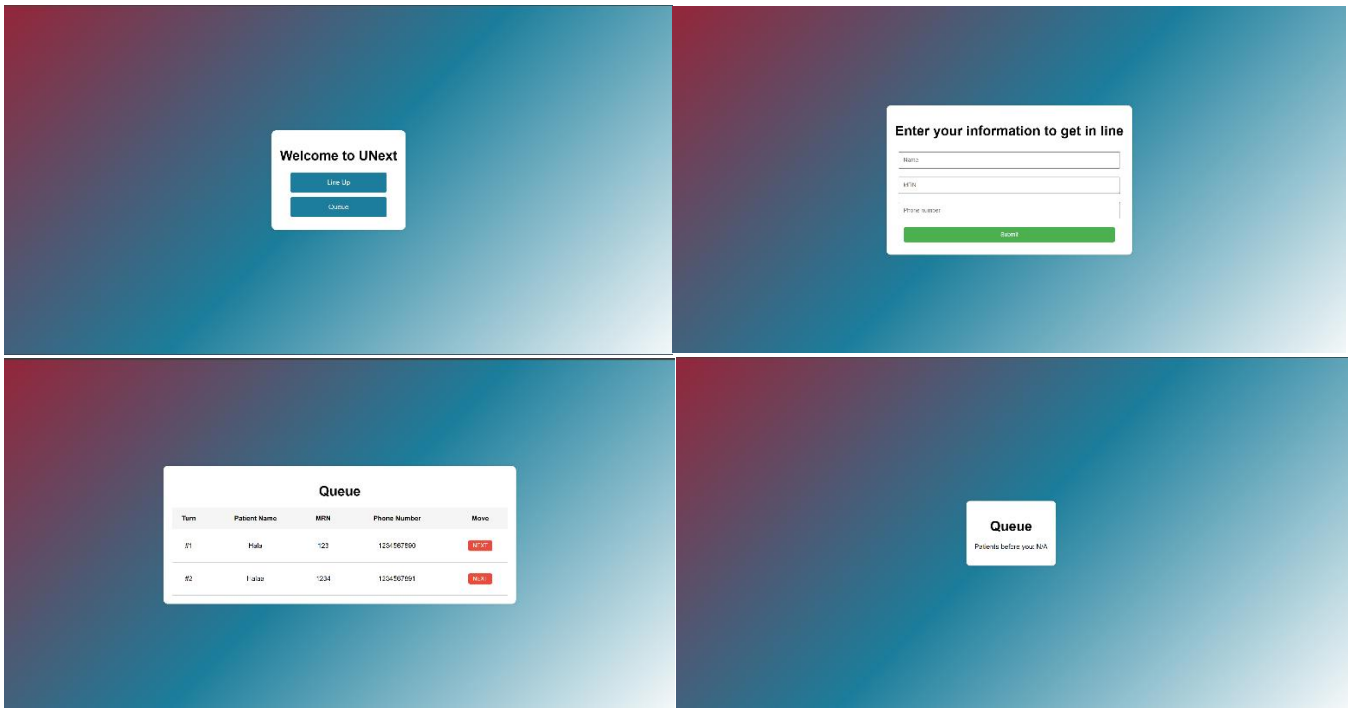
One of the most significant projects I worked on was the **OneClick software**. This project involved developing a system that synchronizes advertisement screens to display the same content simultaneously across different locations. The scope of this project was comprehensive, requiring work from frontend design to backend development. On the frontend, I was responsible for creating and debugging the software's interface using HTML, CSS, and JavaScript. This involved designing a user-friendly interface that would be intuitive for users to interact with.

On the backend, I focused on integrating user registration and login functionalities, as well as enhancing the user experience by implementing features like "change password." This full-stack approach allowed me to develop a deeper understanding of how the frontend and backend interact, ensuring seamless integration and functionality. The project not only allowed me to apply my technical skills but also challenged me to think critically about user interaction, data management, and software integration.

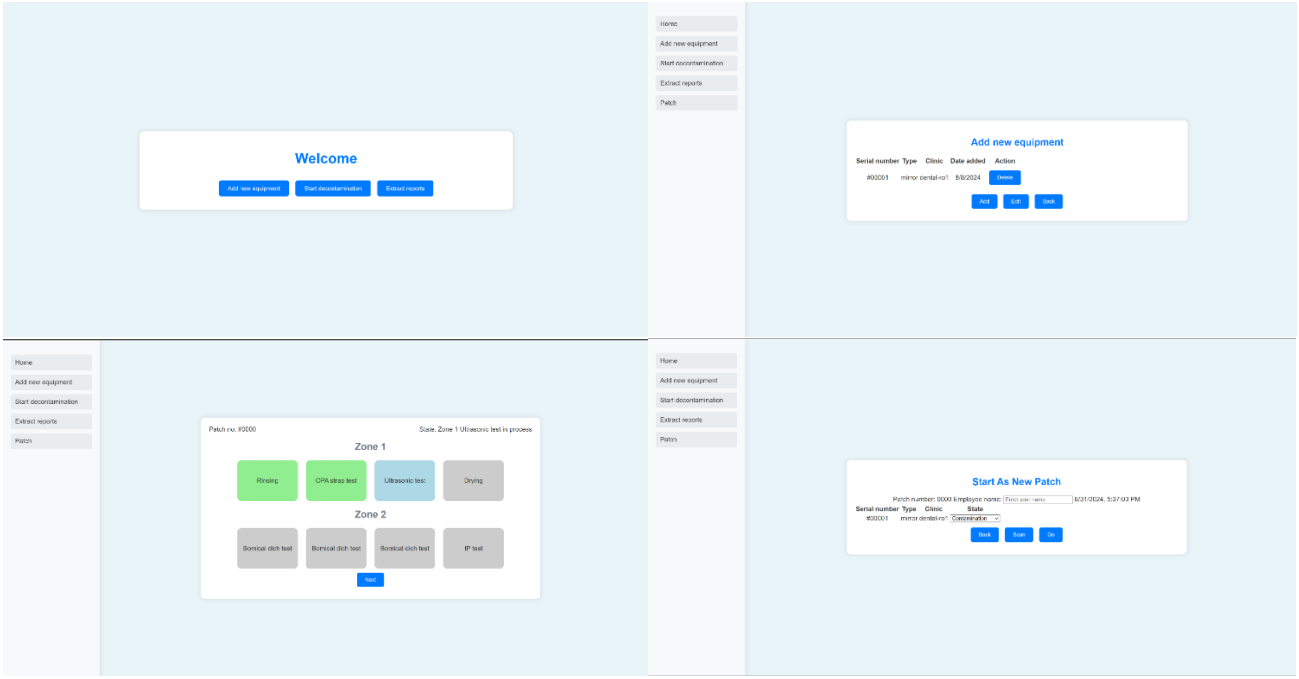




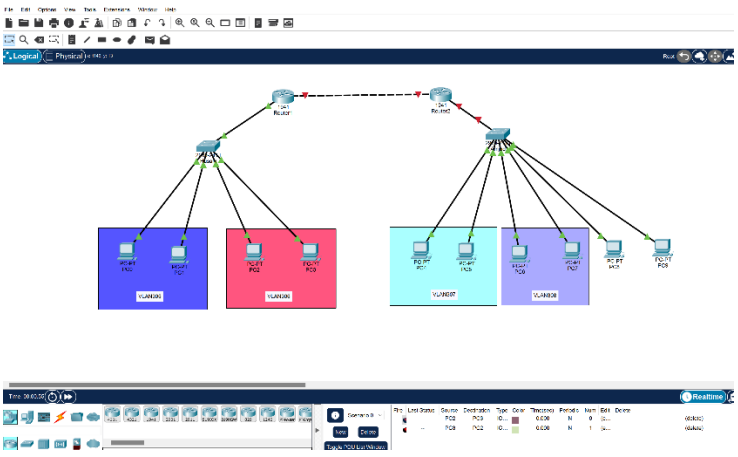
Another project that stood out was the development of a **queuing system named UNext**. This project required designing and developing interfaces that would manage patient queues efficiently. The project was particularly interesting as it involved understanding the flow of patients within the medical facility and translating this into a digital system that could improve patient experience and operational efficiency. Working on UNext provided me with insights into how technology can be used to solve real-world problems in healthcare settings.



I also contributed to the development of a **system for the sterilization department**. This project involved tracking the sterilization processes of medical equipment, ensuring that all equipment is properly cleaned and ready for use. The project required close collaboration with the department staff to understand their workflow and translate it into a functional digital solution. This experience was invaluable in teaching me how to gather requirements, design user-friendly interfaces, and implement effective software solutions.



In addition to software development, I also built a network system. The network consisted of multiple VLANs (Virtual Local Area Networks) connected through switches and routers to ensure secure and efficient communication between different segments of the organization. The design of the network required careful planning and configuration, as each VLAN was assigned isolated ranges to prevent unauthorized access and ensure data integrity. This experience provided me with hands-on knowledge in networking, from setting up VLANs to configuring routers and switches, which was essential for the smooth operation of the OneClick software.



Interesting Events

One notable event was the detailed session with the IT team where we outlined the workflow for the OneClick project. This session was crucial as it involved understanding requirements, documenting VM specifications, and initiating server setup within the VM environment. It was an excellent learning opportunity that provided me with practical experience in project management and system setup.

Another interesting event was the comprehensive tour of the medical facility, where I assessed technical and systemic issues for potential improvements. This hands-on experience was invaluable as it provided a clear understanding of the real-world challenges faced by healthcare facilities and how IT solutions can address these challenges.

Additionally, we held several meetings with the Chief Operating Officer (COO) to discuss the OneClick project. These meetings were essential for aligning the project's objectives with the organization's broader goals. We assessed the technical and systemic issues that needed to be addressed and strategized on how the OneClick system could be optimized to meet these needs. This interaction with senior management offered valuable insights into how high-level decisions impact project implementation and outcomes.

Skills Developed

Throughout my training, I developed several technical and soft skills that will be invaluable in my future career:

Technical Skills

- **Backend Software Development**
  - Enhanced proficiency in creating and managing databases.
  - Integrated various systems to ensure seamless operation.
- **Front-End Development**
  - Improved skills in HTML, CSS, and JavaScript for designing and developing responsive interfaces.
  - Enhanced ability to create visually appealing and user-friendly designs.
  - Developed user-friendly interfaces to improve user experience.
- **Networking**
  - Gained experience in setting up and configuring networks.
  - Learned how to connect devices efficiently and troubleshoot network issues.

Soft Skills

- **Project Management**
  - Learned how to manage multiple projects simultaneously.
  - Improved ability to prioritize tasks and meet deadlines effectively.
- **Time Management**
  - Improved ability to manage deadlines and deliverables within tight schedules.

- **Communication**
  - Developed skills in communicating with people in higher positions.
  - Learned how to effectively explain ideas and viewpoints.
  - Gained experience in negotiating to reach mutually beneficial outcomes.
  - Honed skills in gathering requirements from non-technical stakeholders.
- **Teamwork and Collaboration**
  - Worked closely with different teams, enhancing collaboration skills.
  - Improved teamwork abilities, essential for any professional environment.

Overall, my summer training at Medical Fakeeh was an enriching experience that provided me with practical skills, broadened my understanding of the healthcare IT sector, and prepared me for future challenges in my career.

Challenges

During my summer training at Medical Fakeeh, I encountered several challenges that tested my problem-solving abilities, adaptability, and resilience. These challenges were valuable learning experiences that contributed to my professional growth.

1. **Balancing Multiple Projects**
   - One of the main challenges I faced was managing multiple projects simultaneously. Juggling the development of the OneClick software, the UNext queuing system, and the sterilization tracking system required careful time management and prioritization. It was challenging to ensure that each project received adequate attention and that deadlines were met without compromising the quality of work. This experience taught me the importance of setting clear priorities and managing my time effectively.
2. **Technical Difficulties**
   - Throughout the training, I encountered various technical issues, particularly during the development and debugging phases of the projects. For example, integrating the backend systems with the frontend interfaces in the OneClick project was particularly challenging. There were instances where the software did not function as expected, requiring extensive troubleshooting and debugging. These technical difficulties were frustrating at times, but they also provided an opportunity to deepen my understanding of the underlying technologies and improve my problem-solving skills.
3. **Communicating with Non-Technical Stakeholders**
   - Another challenge was effectively communicating with non-technical stakeholders, including the COO and department staff. Explaining complex technical concepts in a way that was easy to understand and ensuring that their feedback was accurately incorporated into the projects required careful communication and negotiation. This experience highlighted the importance of clear and concise communication, particularly when working with individuals who may not have a technical background.

4. **Collaborative Team Dynamics**
    - Working in a team with diverse backgrounds and varying levels of experience presented its own set of challenges. Coordinating efforts, managing differing opinions, and ensuring that everyone was on the same page required strong teamwork and collaboration skills. There were times when differences in approach or understanding led to delays or rework. However, these experiences helped me develop better collaboration strategies and taught me the importance of flexibility and open communication within a team.

Adapting to New Technologies

5. During my training, I was introduced to several new technologies and tools that I had not previously used, such as specific networking configurations and advanced database management systems. Adapting to these new technologies within the limited time frame of the training was challenging. However, it also pushed me to learn quickly and become more adaptable in the face of new and unfamiliar tools, which is a valuable skill in the ever-evolving field of technology.

Overall, these challenges were integral to my learning experience during the summer training. They not only tested my technical and soft skills but also prepared me to handle similar situations in my future career.

Reflection

What is the most important skill that you learned during your training? Justify your answer.

The most important skill I learned during my training at Medical Fakeeh was **communication**. Effective communication is crucial in any professional setting, but during my training, I realized just how vital it is, especially in a complex and dynamic environment like the healthcare sector. I learned how to articulate my ideas clearly and concisely, particularly when discussing technical concepts with non-technical stakeholders, such as the Chief Operating Officer (COO) and department staff. This skill was essential in ensuring that everyone involved in the projects had a clear understanding of the goals, requirements, and progress.

Additionally, I developed the ability to communicate confidently with individuals in higher positions, which helped me gain valuable insights and support for the projects I was working on. Learning how to negotiate and present my viewpoints effectively allowed me to contribute more meaningfully to the discussions and decision-making processes. This experience has shown me that strong communication skills are not just about speaking clearly but also about listening, understanding different perspectives, and facilitating collaboration.

How do you plan to use the skills developed during your training in the future?

The communication skills I developed during my training will be invaluable in my future career. Whether I am working in a technical role, managing a project, or leading a team, the ability to convey ideas clearly, negotiate effectively, and collaborate with a diverse group of people will be critical to my

success. I plan to use these skills to ensure that projects run smoothly, that everyone involved is on the same page, and that any issues are addressed promptly and effectively.

Furthermore, my experience at Medical Fakeeh gave me a clear understanding of how technologies are implemented in real-world settings, particularly in the medical sector. This insight into the practical applications of technology has deepened my appreciation for the role IT plays in improving healthcare. Moving forward, I intend to leverage both my technical and communication skills to contribute to projects that bridge the gap between technology and critical sectors like healthcare, ensuring that technological solutions are effectively integrated to meet real-life needs.

Conclusion

In conclusion, my summer training at Medical Fakeeh was an invaluable experience that allowed me to apply my academic knowledge in a real-world setting while developing crucial skills that will benefit my future career. Through hands-on involvement in projects like OneClick and UNext, I gained a deeper understanding of how technology can be leveraged to solve practical challenges in the healthcare sector. The opportunity to work on both the frontend and backend of these projects provided me with a comprehensive view of the software development process, while the challenges I faced helped me hone my problem-solving abilities and adaptability.

Moreover, the training exposed me to the realities of the professional world, particularly the importance of effective communication, time management, and teamwork. I leave this experience with a strong foundation of technical and soft skills, as well as a renewed commitment to contributing to the intersection of technology and healthcare. I am confident that the lessons learned and the skills acquired during this training will serve me well as I embark on my professional journey.

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